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Navigating the trade-off between energy, capacity investment, labor, and inventory costs through indirect energy storage for sustainable production

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ABSTRACT

As the share of renewable energy sources increases, electricity prices become more volatile, creating both challenges and opportunities for energy-intensive manufacturing industries. This paper introduces a mathematical optimization model that supports annual production planning with a focus on minimizing total costs associated with energy, labor, inventory, and capacity investments. The core idea is to use storable, energy-intensive products as a form of indirect energy storage, producing them when electricity prices are low and capacity is available. By integrating dynamic labor and energy costs at the shift level, the model enables cost-efficient activation of production shifts and guides capacity investment decisions to enhance production flexibility. The model was evaluated in a practice-oriented use case using electricity price data from Ember Energy across multiple scenarios and through sensitivity analyses. Results demonstrate that selectively activating shifts with lower energy and labor costs can significantly reduce total expenses while maintaining production efficiency. This work contributes to more sustainable industrial practices by aligning operational decisions with cost-efficient and environmentally conscious energy use.

1. Introduction

The energy market is transforming towards renewable energies, leading to variability and volatility in the electricity production volumes and prices (Rintamäki et al., 2017; Lee and Cho, 2022). The unified EU framework for promoting renewable energy set an overall binding target of a 42.5% share of renewables by 2030 (European Commission, 2023). The resulting price volatility leads to several challenges (Chu, 2012; Golmohamadi, 2022). Optimizing energy consumption by utilizing electricity price fluctuations represents a promising path for energy-intensive industries, e.g., basic metals, paper, coke and petrol, chemicals, plastics, minerals, textiles, and wood (Bijnens et al., 2022). However, to minimize energy costs in labor- and energy-intensive industries, several environmental and economic trade-offs arise: energy versus labor (Mazzanti and Zoboli, 2009), asset (Steinder et al., 2008), and inventory costs (Jauhari, 2022). These environmental and economic trade-offs (Tognetti et al., 2015) can be addressed by utilizing production flexibility and appropriately rescheduling energy usage. This enables industrial customers to capitalize on pricing mechanisms through the adoption of critical peak pricing to achieve significant cost savings (Wang and Li, 2014).

To benefit from such approaches, it must be noted that energy pricing systems are dynamic and complex (Wang and Li, 2014). Energy costs depend on energy consumption, but additional charges apply for peak demand. Typically, varying time-of-use rates apply in the industrial context. High electricity demand and low electricity volume production drive the electricity price to temporarily high levels. Simultaneously, renewable energy sources can lead to even negative energy prices during high energy production phases (Csereklyei, 2020). Research shows that an increase in electricity energy prices per kWh decreases energy consumption, reduces CO_2 emissions, and has a relatively small negative impact on employment (Bijnens et al., 2022; Marin and Vona, 2017). Further, energy price impacts are larger in the long run than in the short run (Marin and Vona, 2021; Kwon et al., 2016), except for productivity, as capital deepening exacerbates job destruction but mitigates efficiency losses.

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Thus, energy-intensive production demands smarter and more efficient methods for managing energy consumption, particularly amid dynamic electricity prices and the trade-offs between energy and operational costs. Several studies have proposed various strategies to optimize energy-efficient production. These include methods such as *electricity price forecasting* from short-term forecasting using standard statistical or AI-based models (Contreras et al., 2003; Ugurlu et al., 2018) to medium term forecasting for production scheduling tasks (Windler et al., 2019) and long-term probabilistic forecasting for strategic decision-making (Ziel and Steinert, 2018), where forecasting models are useful for forecasting electricity price fluctuations, enabling better scheduling decisions. *Participating in demand response programs* (Sloot et al., 2022) and using energy sources during periods of lower consumption intensity are key approaches to reducing energy costs. *Shifting energy-intensive processes* to off-peak hours under real-time or time-of-use pricing schemes has proven effective in lowering electricity expenses (Eryilmaz et al., 2019). *Energy-efficient production scheduling* represents the intersection of operations research, industrial engineering and sustainable manufacturing, classified as reactive and active approaches (Gahm, 2016). Gao et al. (2020) provides further granularity to these three approaches based on multiple problem settings, objectives, and energy-related modeling choices. Reactive approaches include adaptive scheduling methods that respond to dynamic electricity prices to optimize energy use (Lee et al., 2017; Karimi et al., 2021a), and active approaches include scheduling and machine-assignment decisions that utilize differences in energy consumption across machines, independent of price variability (Li et al., 2016). *Long-term control methodologies* ensure energy efficiency through optimized maintenance strategies (Brundage et al., 2015). *Integrated inventory models* optimize inventory levels, investments, energy use, carbon taxes, and lot-sizing in supply chain systems (Jauhari, 2022). The approaches discussed have several limitations, as they do not simultaneously consider factors from demand management and scheduling, such as dynamic energy prices, labor, and inventory cost constraints, leading to limitations in energy-efficient production optimization and planning (Gong et al., 2019a; Ranaboldo et al., 2024).

When prices are low or renewable energy production is high, energy storage systems can store excess energy for use during peak demand periods, thereby reducing reliance on the grid (Elio, 2021; Olabi, 2021). However, the batteries of the energy storage system degrade and incur higher costs over time (Lee, 2022). Additionally, energy storage systems pose environmental and sustainability challenges (Dehghani-Sanij, 2019; David, 2021).

An alternative approach to traditional energy storage systems is to indirectly "store energy" by building up product inventory during periods of low electricity prices. By strategically scheduling production and leveraging flexible manufacturing configurations, companies can shift energy-intensive processes away from high-cost periods, thereby achieving substantial energy cost savings, even in comparison to on-site solar power generation and battery energy storage (Karimi et al., 2021b; Peinado-Guerrero, 2022). However, scheduling production activities to produce energy-intensive products as a proxy for energy storage, based on dynamic energy pricing, is not a straightforward approach. It requires consideration of other factors, e.g., production process flexibility, difficulty in adjusting energy-intensive production, supply chain operational dependencies, workforce availability, maintenance challenges, and inventory availability. The integration of these operational factors into energy-flexible production scheduling strategies has received comparatively little research attention. Any attempt to address such challenges without considering the aforementioned factors can lead to suboptimal solutions, resulting in energy-inefficient output as well as production and cost inefficiencies.

Addressing these challenges, the central idea of the article is to use electrical energy to produce energy-intensive raw materials, components, or products for energy-efficient production through optimized energy-aware scheduling, along with consideration of inventory, labor, and investment costs. Therefore, this study develops a mathematical optimization model for yearly production planning to minimize involved costs such as electricity, inventory, labor, and investment—costs that are often overlooked in other studies under similar assumptions. Based on the optimization model, whenever electricity prices are low, energy-intensive processes should be activated. Hence, usable capacity must be available when electricity prices are low, and products should be highly energy-intensive and easily storable. This study not only improves energy-efficient production but also contributes to broader SDGs by minimizing costs that ensure more affordable energy (SDG 7), more sustainable consumption and production (SDG 12), and sustained growth and productive employment (SDG 8).

Hence, this study attempts three main contributions:

1. A novel energy-aware production scheduling approach that directly uses low-cost electricity for producing storable, energy-intensive products instead of storing it in the energy storage system.
2. An integrated optimization model for annual planning that minimizes electricity, inventory, labor, and investment costs.
3. The approach improves sustainable and cost-efficient production, contributing to long-term planning, affordable energy, sustainable production, and economic growth, supporting several SDGs.

In the proposed optimization approach, sales and electricity prices are assumed to be known, or reliable forecasts are available. Investments and labor are included in the model, relating to the trade-off between utilization and time-flexible usage of production capacity.

The paper is structured as follows: Section 2 provides a review of related work on energy cost optimization in production and manufacturing. Section 3 covers dynamic energy pricing, describing characteristics of volatile electricity prices. Section 4 introduces the proposed optimization model. In Section 5, the model is applied to a practical use case to demonstrate its applicability. Section 6 presents the findings and managerial insights, and Section 7 concludes the study.

2. Related work

In this section, we review studies on energy-efficient methods, such as dynamic pricing, scheduling, and optimization, under various scenarios aimed at reducing energy costs and promoting energy-efficient production.

2.1. Dynamic pricing models

The addition of renewable energy sources (RES) to the power grid has drastically changed the energy landscape, leading to a significant increase in the use of dynamic electricity pricing mechanisms for sustainable, renewable energy-aware machine scheduling (Ertem, 2024). The price volatility is the key input and economic driver for the optimization model introduced in this paper. Since prices fluctuate with supply, the time-dependent difference between high and low electricity costs presents a core opportunity to use storable, energy-intensive products as indirect energy storage. Therefore, a *dynamic pricing model*-based studies are important to establish the reality that enables the cost-efficient activation of production shifts and guides capacity investment decisions, which are the primary focus of this optimization study.

Dynamic pricing models adjust prices based on market demand, customer behavior, and other market dynamics, aiming to optimize revenue and thereby achieve greater long-term profitability compared to marginal pricing models (Den Boer, 2015). Various industrial sectors have adopted these models. In transportation, key studies have been performed by Saharan et al. (2020) and Poláček et al. (2024). In the electricity sector, Hu et al. (2015) reviewed dynamic pricing programs, Dutta & Mitra (2017) examined dynamic pricing from revenue management perspective, and Hao et al. (2024) performed a thematic

mapping of the field. In online marketplaces dynamic pricing has been studied by Schlosser & Boissier (2018). This adoption is mainly due to technological advancements that allow companies to exploit data from customer activities and consumer behavior to extract relevant knowledge for dynamic pricing models (Den Boer, 2015). These dynamic models aim to maximize accumulated profit as a key criterion for evaluating different market opportunities (Robinson and Lakhani, 1975).

In the energy sector, demand can be managed based on usage through dynamic pricing—a *demand-side management* technique—thereby reducing peak load and avoiding costly capacity expansions and capital investments (Dutta and Mitra, 2017). Shifting demand from peak to off-peak periods improves system efficiency, cost, and reliability, along with a positive environmental impact and grid reliability.

Demand-side energy management compels consumers to adjust their energy consumption based on real-time price changes for cost savings, known as implicit or explicit industrial demand response (Ranaboldo, 2024). This can be achieved by monitoring real-time pricing, shifting activities to off-peak hours, participating in demand response programs (Sloot et al., 2022), and optimizing energy sources during periods of less intensive energy usage.

For industries, dynamic energy pricing demand response can be used to optimize production activities and reduce energy costs in production. Consumers in manufacturing can estimate the energy prices of peak and off-peak hours using different approaches, including time-of-use pricing plans, real-time pricing, and utilizing data-driven models for forecasting energy prices based on historical market data, supporting demand response. These methods help manufacturers manage energy costs and usage effectively by shifting production activities to periods when prices are lower.

2.2. Production scheduling with energy optimization

Several approaches, including models and algorithms, are applied to optimize *production scheduling* processes while considering energy-efficient production. Energy-efficient production scheduling involves single-objective strategies that pursue energy-related goals, as well as multi-objective strategies that integrate both energy and traditional scheduling goals (Gao et al., 2020). These strategies employ various frameworks and models, including standard mathematical programming models such as integer programming, integer linear programming, mixed integer linear programming, mixed integer nonlinear programming, mixed integer programming, and nonlinear programming. Other non-traditional approaches include feedback control, neural networks, decision support systems, monitoring systems, and time-series models.

The *EnergyBlocks methodology* for planning and operating energy-efficient production systems integrates energy-efficiency criteria into production system planning by segmenting processes into operations with defined energy consumption (Weinert et al., 2011). Mixed integer quadratically constrained programming optimization addresses flexible job-shop problems by accounting for time- and machine-dependent electricity costs and incorporating the distribution of energy resources, such as energy storage and renewable energy sources (Moon et al., 2014).

Li et al. (2016) minimize tardiness and energy consumption for a parallel *machine scheduling* problem in energy-efficient scheduling. They developed a mathematical model and applied ten heuristic algorithms, discovering that combinational rules were most effective in this context. The multi-objective energy- and *labor-aware production scheduling* model proposed by Gong et al. (2019b) incorporates real-time electricity pricing and is solved using an adaptive multi-objective memetic algorithm. The approach aims to minimize makespan as well as combined energy and labor costs, subject to constraints related to price-aware energy consumption, labor, jobs, changeovers, and machines.

Gong et al. (2019a) highlight *energy scheduling optimization* and present a flexible job shop scheduling model with multiple objectives that integrates energy cost and workforce considerations. This model addresses the challenge of balancing electricity prices with labor wages during scheduling by optimizing makespan, total energy cost, total labor cost, maximum workload, and total workload. The model adopts a nondominated sorting genetic algorithm for multi-objective optimization.

Catanzaro et al. (2023) review studies on rescheduling problems in manufacturing under *time-of-use-based tariffs* aimed at achieving sustainable outcomes. They provide a comprehensive classification of job scheduling problems, including single-machine, parallel-machine (identical and unrelated), flow shop, and job shop scheduling, based on both single- and multi-objective optimization approaches. The review highlights the challenges in scheduling optimization, particularly the limitations in the scalability and effectiveness of existing approaches for multi-machine scheduling, which require the adoption of faster approximation or exact algorithms.

Ranaboldo et al. (2024) present a comprehensive review on scheduling problems addressed through heuristic optimization at the operational level, including single-machine, parallel-machine, flow-shop, and job-shop scheduling.

2.3. Exploitation of dynamic electricity prices

A cement plant reduced electricity costs by shifting the electricity load of certain processes to periods of lower electricity prices. Additionally, it decreased electricity use during peak pricing through more efficient production scheduling under real-time pricing compared to fixed pricing (Eryilmaz et al., 2019). Jauhari (2022) introduces an *inventory model* to optimize inventory levels, investments, and energy consumption in a manufacturer-multi-retailer system, addressing carbon tax policy and lot-sizing strategies.

Duarte et al. (2020) optimize operations scheduling and energy management system decisions by incorporating uncertainty in renewable energy source power output and energy supply under demand-side management policies for multi-process production scheduling. They propose a two-stage mixed-integer programming framework: in the first stage, the production schedule is planned over a given time horizon with associated costs; in the second stage, the model assumes a worst-case RES output scenario, and the energy management system adjusts operations in real time to minimize energy-related cost.

Karimi et al. (2021a) explored the application of *time-of-use energy tariffs* in additive manufacturing by strategically adapting the production schedule, selecting process parameters, and integrating them as components of the scheduling decision to optimize energy costs and effectively level peak power demand.

Burmeister et al. (2024) discuss the *multi-objective optimization* to model the trade-off between minimizing makespan and energy pricing for energy cost-aware scheduling. They formulate a multi-objective flexible job shop scheduling problem (FJP) with dynamic energy costs and propose a memetic evolutionary algorithm for optimization.

Other than scheduling with energy cost savings, a recent study by Dunke et al. (2025) examines energy-aware job shop scheduling as a multi-objective optimization approach to minimize makespan, energy costs, energy consumption, and carbon emissions through a mixed-integer nonlinear programming model. This model accounts not only for machine-specific energy consumption and the use of both conventional and renewable energy sources, but also for diverse tariff structures that incorporate dynamic pricing, energy-dependent processing characteristics, and power purchase agreements.

2.4. Research gap

The articles mentioned above address specific, isolated aspects such as dynamic pricing, scheduling, and energy optimization. However, none explicitly focus on using energy-intensive production components

as a form of energy storage by leveraging dynamic electricity prices in combination with excess capacity and shift-dependent labor costs. This article addresses this gap by introducing a production scheduling model that minimizes the total costs associated with electricity, labor, inventory, and investment.

3. Dynamic energy pricing

Electricity prices vary over time, with both short-term and long-term fluctuations reported. Matisoff et al. (2020) report a strong positive correlation between county-wide energy demand and dynamic electricity prices in the U.S. electricity market, which reach their daily peak at about 6 pm. Wang et al. (2019) report electricity prices varying between 17 $/MWh and 24 $/MWh (about 40%) for the investigated U.S. company. Heidarpanah et al. (2023) state even higher daily fluctuations, ranging between 1.3 $/MWh and 2.1 $/MWh (about 60%) for the Iranian electricity market. Some authors assume short-term (e.g., every 15 min) fluctuations in their optimization models. For example, Lee et al. (2017) use price changes between 0.025 $ and 0.1 $/kWh over 90 min.

According to Bijnenes et al. (2022), average electricity prices in Europe vary between countries, ranging from about 0.06 €/kWh (Sweden) to about 0.15 €/kWh (Italy) in 2016. Ugurlu et al. (2018) developed a forecast model for daily average electricity prices, ranging from about 0 € to 70 €/MWh. Contreras et al. (2003) built an ARIMA model to predict weekly forecasts with hourly time discretization, spanning from about 15 €/MWh to 45 €/MWh. Vattenfall (https://www.vattenfall.de/strom/tarife/oekostrom-dynamik-boersenpreise) offers a public service that shows hourly dynamic stock electricity prices. These prices cover historical and current data. Typically, in February 2025, there are two daily peaks at 8 a.m. and 5 pm., and two troughs at 2 a.m. and 2 pm. The hourly prices range from about 0.10 to 0.16 €/kWh at their troughs and from about 0.25 to 0.30 €/kWh at their peaks. Similar high fluctuations occur in other months. Electricity price information is also available at https://ember-energy.org/data/european-wholesale-electricity-price-data/. Ember provides past hourly electricity prices for all European countries. The German market data for 2024 will be used in the case study presented in Section 5. Based on the data provided by Ember, the average electricity prices for the day shift (6 a.m. to 2 pm.), late shift (2 pm. to 10 pm.), and night shift (10 pm. to 6 a.m. the next day) for all days in the year 2024 were calculated. Table 1 shows the high fluctuations in these average prices.

These high price fluctuations promise some optimization potential for highly energy-intensive production processes. Dutta and Mitra (2017) present an overview of several dynamic electricity pricing policies, including block rate tariffs, seasonal tariffs, time-of-use tariffs, and super-peak tariffs.

This study's optimization model considers all these fluctuations: average long-term variations, day-to-day variations, daily variations, and shift-related variations. Production systems are organized in time shifts, typically in 8-h shifts. For an optimization problem considering investment decisions and the trade-off between inventory, personnel, and energy costs, it makes no sense to subdivide the 8-h shifts, which are thus used for modeling the daily and yearly price fluctuations. The prediction of the average electricity prices for each shift for one year was based on past data. Models such as those presented by Ugurlu et al. (2018) or Contreras et al. (2003) can be used for this prediction. Furthermore, analyzing past daily price variations (Wang et al., 2019; Heidarpanah et al., 2023), the average prices per shift can be estimated.

Table 1
Fluctuations of average shift electricity prices in ct/kWh.

Price	All shifts	Day Shift	Late Shift	Night shift
Minimum price	−0.0510	−0.0510	0.0089	−0.0013
Maximum price	0.4762	0.4641	0.4762	0.1329
Average price	0.0777	0.0630	0.0978	0.0725

The year is divided into working days and working days into shifts. Based on forecasts and past daily price distributions, a shift-dependent average electricity price is assumed for each shift.

4. Model development

An optimization model is proposed that determines which shifts should be activated within a planning period, which production program should be executed, and which additional machine capacity should be newly invested in at the beginning of the considered time frame. Only in activated shifts can the production of multiple products take place. The model incorporates inventory relationships, machinery constraints, and personnel capacity limitations, aiming to minimize all relevant costs, including those related to inventory, energy, labor, and investment.

4.1. Notation and time model

We consider a production system that manufactures multiple products using various types of machinery and workers with different personnel qualifications. Index i $i(i=1,\ldots,m)$ refers to machinery i, index $j(j=1,\ldots,n)$ to product j, and index $l(l=1,\ldots,p)$ to personnel qualification group l. m is the total number of used machineries, and n is the total number of considered products (finished goods, intermediate parts, components, raw materials). For modeling time periods, the index $t(t=1,\ldots,T)$ is used to refer to the $t-$ th shift during the year. T is the total number of shifts per year. Each day of the year is divided into three shifts, regardless of whether it is a working day or a holiday:

- Day shift (6 a.m. to 2 pm.)
- Late shift (2 pm. to 10 pm.) and.
- Night shift (10 pm. to 6 a.m. next day)

Table 2 provides an overview of the variables and parameters used for modeling relationships in the following subsections.

4.2. Modeling shift-independent relationships

In the first step, relationships that are independent of the specific shift are defined. Jodlbauer and Altendorfer (2010) and Kerkhove et al. (2017) address the trade-off between investment, labor, and inventory costs. To model these trade-offs, this study's model distinguishes between machine capacity (including tools, processes, and devices) and personnel capacity. At the beginning of the planning period, investments in machine capacity are allowed and remain available throughout the entire time frame under consideration (Martínez-Costa et al., 2014; Jodlbauer, 2024). If a shift is activated, its available capacity must be sufficient to produce the production program x:

$$Ax \leq h + Qg \tag{1}$$

On the left side of inequality (1), the required machine capacity Ax to produce the program x in an activated shift is represented. On the right side, the available machine capacity is split into pre-existing capacity h and newly invested capacity Qg. The machine threshold matrix Q is diagonal, with its diagonal elements describing the additional usable capacity provided by each newly invested machine for one activated shift (usually 8 h).

The required personnel capacity b to exploit the total machine capacity $h + Qg$ in an activated shift is defined by:

$$b = B(h + Qg) \tag{2}$$

Additionally, the energy consumption required for an activated shift must be considered. The electricity e needed to operate the full machine capacity during an activated shift is represented by:

$$e = r^T(h + Qg) \tag{3}$$

Table 2
Overview of used symbols, variables, and parameters.

Variable	Description	Type
m	number of machineries	input
n	number of products	input
p	number of personnel qualification groups	input
T	number of shifts	input
$\bar{z}_t \in \{0,1\}$	binary shift activation variable, which determines if shift t is activated (1) or not (0)	decision variable
$x = (x_j) \in \mathbb{R}_n$	number of products j produced in an activated shift	decision variable
$g = (g_i) \in \mathbb{N}_m$	number of invested machines of type i	decision variable
$Q = \left(q_{ij}\right) \in \mathbb{R}_m^m$ diagonal matrix	q_{ii} usable capacity for one newly invested machinery of type i	input
$A = (a_{ij}) \in \mathbb{R}_m^n$	capacity needed from machinery i to produce one unit of material j	input
$h = (h_i) \in \mathbb{R}_m$	available machinery capacity of machinery i for one shift	input
$B = (b_{lj}) \in \mathbb{R}_p^m$	capacity needed from personnel qualification group l to run machinery j in one shift	input
$b = (b_l) \in \mathbb{R}_p$	required personnel capacity of qualification group l for an activated shift	auxiliary variable
$r = (r_j) \in \mathbb{R}_m$	energy consumption for one capacity unit of machinery j	input
e	required energy to run all machinery in an activated shift	auxiliary variable
$P = \left(p_{jk}\right) \in \mathbb{R}_n^n$	quantity of needed products j to produce one unit of product k ($p_{jj} = 0$) (bill of material matrix)	input
$y_t = \left(y_{t,j}\right) \in \mathbb{R}_n$	inventory of product j at the end of shift t (= to the beginning of shift $t+1$)	auxiliary variable
$y_{\text{initial}} = \left(y_{\text{initial},j}\right) \in \mathbb{R}_n$	initial inventory of product j	input
$y_{\text{end}} = \left(y_{\text{end},j}\right) \in \mathbb{R}_n$	end inventory of product j	input
$s_t = (s_{t,j}) \in \mathbb{R}_n$	sale of product j during shift t	input
$c_{\text{invest}} = \left(c_{\text{invest},i}\right) \in \mathbb{R}_m$	annual deprecation for a new machinery of type i	input
$c_{\text{inventory}} = \left(c_{\text{inventory},j}\right) \in \mathbb{R}_n$	inventory cost for one unit of material j stored for one shift	input
$c_{\text{energy},t}$	unit energy cost in shift t (average electricity price for the shift)	input
$c_{\text{staff},t} = \left(c_{\text{staff},t,l}\right) \in \mathbb{R}_p$	labor cost for one hour per employee in qualification group l and shift t	input

4.3. Modeling shift-dependent relationships

Electricity prices, unit personnel costs, sales, and inventory are modeled on a shift-specific basis. The index t refers to the t −th shift within a year, with a total of T shifts modeled. The key binary decision variable z_t indicates whether a shift is activated ($z_t = 1$) or not ($z_t = 0$). For an activated shift, the available capacity $h + Qg$ is utilized to execute the production program x. The production program x encompasses all relevant items, including finished goods, intermediate parts, components, and raw materials.

Furthermore, an activated shift requires personnel capacity b and energy consumption e (see Eq. (2) and Eq. (3)). Consequently, shift-dependent labor costs can be calculated by ${c_{\text{staff},t}}^T b$ for an activated shift, while shift-dependent energy costs are determined by $c_{\text{energy},t} e$.

Sales values, respectively forecasts, are known in advance and are not a decision in the optimization problem. The bill of material matrix P specifies the direct material requirements (e.g., components, raw materials). The inventory equation is derived from the production volume driven by fixed sales, the bill of materials breakdown, and existing inventory levels (see Jodlbauer, 2005; Jiang and Rim, 2016).

$$y_t = y_{t-1} + z_t x - P z_t x - s_t \tag{4}$$

The inventory balance Eq. (4) can be interpreted as a generalized discrete form of the net-output process equation introduced by Sigman and Yao (1994). This equation determines the inventory y_t at the end of the shift t as the inventory y_{t-1} at the beginning of shift t (which is equal to the inventory at the end of shift $t-1$) plus the produced material $z_t x$ minus the internal required material $P z_t x$ to produce the production program $z_t x$ and minus sales s_t. Regarding the nonlinear term $z_t x$: This term represents the number of produced items for in-house manufactured parts and the number of sourced items for purchased material during an activated shift ($z_t = 1$). In contrast, when a shift is not activated ($z_t = 0$), no production or purchasing occurs.

The condition of non-negative inventory (Jodlbauer, 2006) ensures that at the end of each shift, all sales requests s_t are fulfilled.

$$y_t \geq 0 \tag{5}$$

Hishamuddin et al. (2020) show the importance of inventory boundary conditions in enabling rolling planning and continuous overlaps between several planning horizons. Boundary conditions for the inventory specify the initial inventory and the targeted inventory at time T.

$$\begin{aligned} y_0 &= y_{\text{initial}} \\ y_T &= y_{\text{end}} \end{aligned} \tag{6}$$

4.4. Objective and model summary

To complete the model, we formulate the objective function. As discussed above, for each shift, shift-dependent average electricity and unit personnel costs are assigned. Based on the approach of Tognetti et al. (2015), the goal is to minimize all associated costs, which can be influenced by decisions concerning investments (new machines), the activation of shifts, and the production program that drive energy, labor, and inventory costs. Energy unit costs (electricity prices) and personnel capacity unity costs are assumed to be shift-dependent. Shift-dependent unit staff costs are modeled as follows: For the day shift on a normal weekday, the standard unit cost for one hour is applied. For the late and night shifts, the unit cost is increased by an appropriate surcharge factor. Furthermore, for public holidays and weekends, the unit cost is further increased. All other unit costs are shift-independent.

The total cost C_{total} for the considered time period comprises shift-independent investment costs, the sum of inventory costs across all shifts t, and shift-dependent energy and personnel capacity costs only for activated shifts:

$$C_{\text{Total}} = {c_{\text{invest}}}^T g + \sum_{t=1}^{T} \left({c_{\text{inventory}}}^T y_t + z_t \left(c_{\text{energy},t} e + {c_{\text{staff},t}}^T b \right) \right) \tag{7}$$

Putting everything together, we obtain the following nonlinear optimization problem:

$$\begin{aligned} \underset{x,g,z_t}{\text{Min}} \; & {c_{\text{invest}}}^T g + \sum_{t=1}^{T} \left({c_{\text{inventory}}}^T y_t + z_t \left(c_{\text{energy},t} e + {c_{\text{staff},t}}^T b \right) \right) \\ & x \geq 0 \\ & z_t \in \{0,1\} \text{ for all } t = 1, \ldots, T \\ & g \in \mathbb{N}_m \\ & Ax \leq h + Qg \\ & b = B(h + Qg) \\ & e = r^T (h + Qg) \\ & y_t = y_{t-1} + z_t x - P z_t x - s_t \text{ for all } t = 1, \ldots, T \\ & y_t \geq 0 \text{ for all } t = 1, \ldots, T \\ & y_0 = y_{\text{initial}} \\ & y_T = y_{\text{end}} \end{aligned} \tag{8}$$

The variables $x \in \mathbb{R}_n$, $g \in \mathbb{N}_m$ and $z_t \in \{0,1\}$ must be determined by solving the nonlinear optimization problem (8). Staff requirement $b \in \mathbb{R}_p$, the energy requirement $e \in \mathbb{R}$ for an activated shift, and the inventory $y_t \in \mathbb{R}_n$ serve as auxiliary variables; all other parameters are

predefined inputs.

For an activated shift, inventory, labor, and energy costs are incurred. The total labor and energy cost per shift can vary significantly. The optimization model (8) aims to activate the cheapest shifts while deactivating the most expensive ones in terms of labor and energy, ensuring non-negative inventories and accounting for inventory costs. Investing in additional machine capacity increases investment costs but also expands the available capacity per shift. This allows for the activation of more cost-efficient shifts while reducing reliance on higher-cost shifts in terms of labor and energy expenses.

The model's conditions (1), (2), (3), (5), and (6) are linear. However, the inventory balance Eq. (4) and the objective function are nonlinear. The multiplication of two different (auxiliary) variables ($z_t x$, $z_t e$, and $z_t b$) causes nonlinearity. This specific type of nonlinearity is called multilinearity because each of the three products is separately linear for each individual (auxiliary) variable. The three auxiliary variables y_t, b and e depend on the investment variable g according to Eqs. (1), (2), and (3). An increase of the inventory y_t is triggered by the production program x in an activated shift, which is restricted by (1). The two auxiliary variables $b$and$e$ are multiplied by the variable z_t in the objective function. The required variables are real ($x \in \mathbb{R}_n$), integers ($g \in \mathbb{N}_m$), and binary ($z_t \in \{0, 1\}$). Model (8) is a mixed integer multilinear programming problem and there are some problem-specific solver engines for that problem class, see Paatero (1999). To solve the model (8), we manually combined the Premium Solver Platform from FrontlineSolvers and a heuristic.

The central idea of the applied heuristic is a forward/backward rolling approach, where the cheapest shift is always activated to avoid the next stock-out. The shift cost is determined by energy, labor, and the incurred holding cost. Although a commercial solver is used, the main methodological contribution of this study lies in the proposed rolling heuristic and its problem-specific cost-based shift activation rule.

5. Use case

In this section, we formulate a practical use case and conduct a numerical study in Excel using Visual Basic programming to evaluate the model developed above, thereby enabling a thorough discussion of the trade-offs between energy costs, capacity investment costs, and labor costs in energy-intensive production. For practical application, only the most relevant products, processes, and machines concerning high energy and labor intensity are considered. Furthermore, similar products are combined in product groups, similar machines in machine groups, and similar personnel qualifications into qualification groups. Additionally, energy price data is needed based on production shifts to enable the optimization of shift-activation or shift-deactivation decisions. The use case presented is kept relatively simple to ensure a straightforward discussion.

5.1. Energy price modeling

To model the energy costs per shift, the investigated use case utilizes the electricity price data for the year 2024 from the German market (available at https://ember-energy.org/data/european-wholesale-electricity-price-data/). Energy prices are available on an hourly basis and are accumulated to calculate the average energy price for individual production shifts. The detailed data shows, first, a strong dependence on energy prices within a daily cycle, and weekends/holidays lead to short-term disturbances in energy prices. Second, seasonality is also observable, as energy is more expensive in the winter season compared to the summer season. Fig. 1 shows an example of energy costs for 21 shifts (i.e., one week) in the summer season and in the winter season.

The data in Fig. 1 shows that energy costs during the night shift are usually lower, and weekends have lower energy costs. Further, in the winter season, the average energy costs are higher, and in the summer season, energy costs per shift might even be negative. This high fluctuation of energy prices is used as input for optimization in the practical use case. Note that Fig. 1 is just an excerpt of the entire data stream applied. The whole data stream consists of $366 \times 3 = 1098$ data points for energy prices per shift.

5.2. Production system structure

To ensure a streamlined discussion, the presented use case considers a simple product structure, i.e., one product group of finished products P1 is produced, which requires, on average, 2.3 pieces of the single component product group P2. On every weekday that is not a public holiday, a constant sale of 900 pieces P1 is assumed. The materials are produced on three machine groups: M1, M2, and M3. Furthermore, two personnel qualification groups, H1 and H2, are modeled. The following factors for the shift-dependent labor surcharge are used: 1.1 for the late shift, 1.2 for the night shift, and 1.25 for public holidays and weekends. These factors must be combined (e.g., for the night shift starting on a public holiday, the factor equals 1.5) and are applied to the unit standard labor cost for a normal working day shift. The inventory holding cost for one piece per shift is 1.3 for P1 and 0.3 for P2. The investigated production system is highly energy-intensive, especially machine 3. Model (8) is applied to the entire year of 2024 with $T = 1098$ available

Table 3
Machine-related input data in the use case.

	Machine capacity A		Available capacity	Energy consumption	Capacity cost	Usable added capacity
Machine	P1	P2	h	r	c_{invest}	q_{ii}
M1	0.1	0.4	310	35	833.33	8
M2	2	0	650	53	2500	8
M3	0	2.1	1600	278	300	8

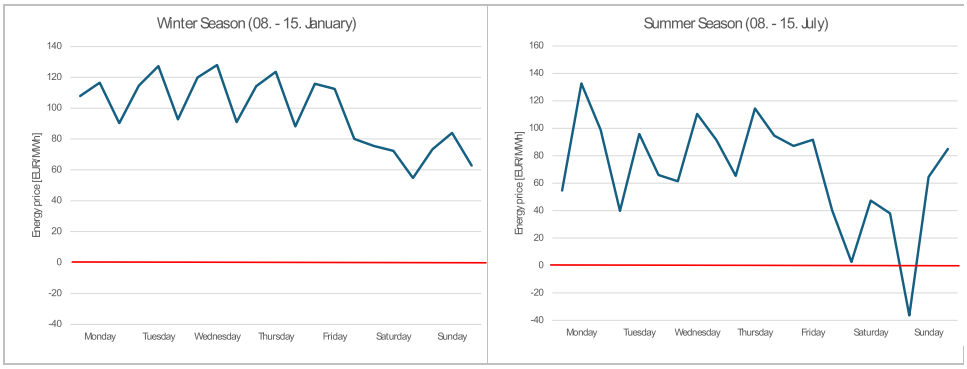


Fig. 1. Example of energy price data applied.

shifts. Tables 3 and 4 provide a detailed overview of the additional input data applied in the use case.

5.3. Scenario definition

To evaluate the potential of aligning the production of energy-intensive products with the energy price, the production and sales system structure introduced above is tested in several scenarios. In Scenario I, the baseline is defined without any optimization. In this scenario, production is conducted in three shifts on each working day. Note that no production occurs on holidays, and therefore, the non-negativity constraint of the optimization problem is violated in this situation. However, the same production program x as in Scenario II is produced in Scenario I, and the same number of shifts is provided to enable a fair comparison. Note that, for synchronizing Scenario I and II, as well as for reaching approximately zero inventory in the final period T, a few shifts at the end of the year are deactivated in Scenario I. The effect of activating and deactivating shifts on weekends and holidays (with higher personnel costs there) is evaluated in Scenario II, but no additional capacity investment is considered. In Scenario III, the full range of decisions is allowed to also discuss the trade-off between energy costs, labor costs, and capacity investment.

To evaluate the effect of capacity investment costs on this trade-off, an additional sensitivity analysis is conducted, where higher capacity costs are tested. Table 5 summarizes the tested scenarios.

5.4. Numerical results for scenarios I to III

To discuss the effect of different decisions on the optimal production system behavior, the results for Scenarios I to III are presented together, followed by a detailed discussion of each scenario. Table 6 shows the shift-dependent decisions and the shift-dependent results.

Since no optimization is performed in Scenario I, it shows the baseline performance of the production system. The production program is defined to fulfill the demand on average; however, due to some holidays, the non-negativity of the inventory constraint is violated in some periods. Nevertheless, the results in Table 6 show the personnel and machine capacity needed when no further decisions can be made. Looking at the energy consumption results in Table 6 reveals the dominating energy consumption of machine 3 in comparison to the other machines. Since no capacity investment decision can be made in Scenario II either, the results of Scenario I and II are the same in Table 6. The results for Scenario III in Table 6 already show what happens when the capacity decision is taken into account. The results show that in Scenario III, the available machine capacity is about 3.5 times higher than in Scenario I and II. This implies that only 209 shifts need to be activated, and the cost reduction potential of activating only cheap shifts (either related to energy costs and/or related to personnel costs) can be exploited here. Table 7 shows the shift-dependent costs for all scenarios, whereby the Min (minimum value), Max (maximum value), and Average (average over all possible shifts) in Table 7 refer to all shifts independently, whether activated or not. The high fluctuations in labor and energy costs per shift are remarkable. These are linked to the energy prices already reported in Fig. 1 above. In all scenarios, labor costs per activated shift vary by a factor of 1.5 (from a day shift on a weekday to a night shift on a public holiday or weekend), caused by the labor surcharge factors. For Scenarios I and II, energy costs per activated shift vary from the negative value −25,007 € to 233,394 €, with the average energy cost per shift equal to 38,130 €. The absolute range of Scenario III for these numbers is even higher (from −81,891 € to 764,312 €); however, this is just related to the higher machine capacity available per shift. The labor and energy costs per shift for Scenarios I and II fluctuate between −7696 € and 252,135 €, with a mean of 58,344 €. Again, the fluctuation in Scenario III is higher based in the higher machine capacity available per shift (from −25,063 € to 826,823 €). For Scenario I, the average labor cost per activated shift of 18,742 € is lower than the average labor cost across all shifts, which is 20,214 €, while the average energy cost per activated shift of 42,475 € is higher than the average energy cost across all shifts, which is 38,130 €. Thus, not surprisingly, on average, labor costs on weekdays are lower than at weekends or public holidays, and more interestingly, energy costs on weekdays are higher than on weekends and public holidays. Looking at these results for Scenario II shows that

Table 4
Personnel capacity matrix B in the use case.

	M1	M2	M3
H1	0.1	0.1	0.05
H2	0.08	0.05	0.05

Table 5
Scenario definition.

Criteria	Scenario I	Scenario II	Scenario III	Sensitivity analysis
Activation of weekend/holiday shifts	no	yes	yes	yes
Capacity investment decision	no	no	yes	yes
Applying different capacity costs	no	no	no	yes

Table 6
Shift-dependent decisions and results.

Shift-dependent decisions and results	Scenario I	Scenario II	Scenario III
Production program per shift in number of items for P1	304	304	1 081
Production program per shift in number of items for P2	699	699	2 487
Staff per shift in number of workers for H1	176	176	588
Staff per shift in number of workers for H2	137	137	457
Machine capacity per shift in hours for M1	310	310	1 103
Machine capacity per shift in hours for M2	650	650	2 162
Machine capacity per shift in hours for M3	1 600	1 600	5 222
Energy consumption per shift in kWh for M1	10 850	10 850	38 599
Energy consumption per shift in kWh for M2	34 450	34 450	114 607
Energy consumption per shift in kWh for M3	444 800	444 800	1 451 763
Number of activated shifts	**744**	**744**	**209**

Table 7
Shift-dependent costs.

Shift-dependent costs	Scenario I	Scenario II	Scenario III
Labor shift cost in € (Min)	17 038	17 038	56 829
Labor shift cost in € (Max)	25 557	25 557	85 243
Labor shift cost in € (Average for all possible shifts)	20 214	20 214	67 422
Labor shift cost in € (Average for activated shifts)	18 742	19 890	67 603
Energy shift cost in € (Min)	−25 007	−25 007	−81 891
Energy shift cost in € (Max)	233 394	233 394	764 312
Energy shift cost in € (Average for all possible shifts)	38 130	38 130	124 866
Energy shift cost in € (Average for activated shifts)	42 475	31 338	49 601
Labor and energy shift cost in € (Min)	−7 969	−7 969	−25 063
Labor and energy shift cost in € (Max)	252 135	252 135	826 823
Labor and energy shift cost in € (Average for all possible shifts)	58 344	58 344	192 288
Labor and energy shift cost in € (Average for activated shifts)	**61 217**	**51 228**	**117 204**

the average labor costs for activated shifts are higher than in Scenario I, meaning some of the shifts on holidays and weekends are activated in this scenario. However, the most expensive shifts related to energy costs are deactivated in Scenario II, which leads to significantly lower energy costs for activated shifts (42,475 € for Scenario I compared to 31,338 € in Scenario II). The average labor and energy costs reported in Table 7 for Scenarios I and II show the overall positive effect of shifting work to shifts with lower energy costs while accepting slightly higher labor costs. The results for Scenario III even show a much higher difference between average energy costs for all shifts (124,866 €) and the average energy costs for the activated shifts (49,601 €). However, this high reduction potential must be traded off with the additional capacity costs, which are shown in Table 7.

The resulting cost performance of the respective scenarios is presented in Table 8, highlighting the cost advantages associated with the individual decision options in Scenario II and Scenario III.

The detailed results in Table 8 show that the flexible use of shifts on weekends and holidays (Scenario II) already implies a very high cost reduction potential in this use case example, i.e., from 47.7 Mio € to 39.4 Mio €. If additional capacity can also be invested (Scenario III), the cost reduction potential even increases, i.e., from 47.8 Mio € to 31.3 Mio €. Looking at the results of Scenario III in detail shows that producing only in shifts with low energy prices leads to energy costs of 32% related to the baseline. However, this is linked to significantly higher machine capacity, which also leads to significant costs, i.e., 4.1 Mio € additional capacity investment costs. Furthermore, inventory costs also increase significantly since utilizing fewer shifts implies a need for pre-production on stock.

5.5. Numerical results sensitivity analysis

Since the good performance of Scenario III is related to investing in high additional machine capacity, a sensitivity analysis is conducted showing the effect of higher capacity cost. Fig. 2 shows the overall costs as well as the individual cost shares for capacity investment cost factors from 100% (Scenario III) to 600%, and Scenario II (infinite capacity cost factor) for comparison.

Comparing the results for the 100% capacity cost factor and the infinite capacity cost factor in Fig. 2 shows the above-mentioned cost reduction potential of 8.1 Mio €. Comparing capacity cost factor = 100% (Scenario III) with capacity cost factor = 200% shows that this increase in capacity costs significantly reduces the cost reduction potential to 5.3 Mio €, and the capacity investment costs decrease from 4.1 Mio € (100%) to 3.4 Mio € (200%). Looking at the numbers in Fig. 2 in detail shows that further increasing the capacity cost factor decreases the overall capacity investment costs. On the other hand, energy costs increase when the capacity cost factor is higher, which is intuitive since less flexibility to change between shifts is available. An interesting finding is that even for a 600% capacity cost factor, a cost reduction potential of 3.3 Mio € is observed compared to Scenario III.

6. Discussion and managerial insights

6.1. Theoretical insights and challenges

Real-world production planning problems are inherently complex due to the vast number of products, processes, and uncertainties in manufacturing processes, demand, and environmental conditions. Multiple factors must be considered when optimizing production planning and control, such as inventory management, operational costs, workforce and machine constraints, and sustainability goals. While the existing literature focuses on optimizing only a subset of these aspects, this study proposes an integrated model that balances energy, labor, and inventory costs. By strategically activating production shifts and making targeted investments in machine capacity, the model enhances flexibility and cost-efficiency. The core concept is to treat energy-intensive components as a form of energy storage—producing them during periods of low electricity prices. This approach, combined with excess capacity and shift-dependent labor costs, allows for dynamic and cost-effective production scheduling.

Its applicability is evaluated in a practice-oriented use case, and several scenarios have been discussed. Especially in energy-intensive manufacturing processes, there is high potential for cost savings. For real-world applications, there are several challenges that must be addressed. To reduce complexity, the focus should be on the most energy- and labor-intensive products, processes, and machines. Additionally, similar products, machines, and personnel qualifications should be grouped to simplify modeling and optimization. Another key challenge is the availability of electricity price forecasts. While precise price predictions may not always be feasible, it is sufficient to classify shifts as either cost-efficient or expensive based on the combined energy and labor expenses. This simplification ensures practical applicability without requiring perfect price forecasts, making the model more robust for real-world decision-making.

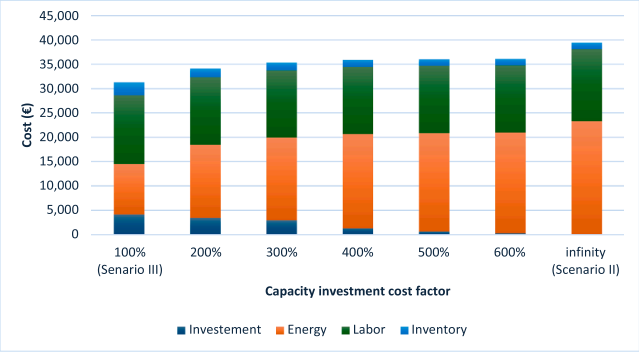


Fig. 2. Cost comparison sensitivity analysis.

6.2. Managerial implications

From a managerial perspective, the findings provide several strategic insights for energy-intensive production systems. First, there is significant potential to activate production on weekends and public holidays to minimize total costs. Since energy prices and labor costs fluctuate over time, adjusting production schedules to take advantage of lower-cost periods can lead to substantial savings. Second, investing in additional machine capacity enhances flexibility in shift scheduling. While this requires upfront capital, it enables companies to allocate production to cost-efficient shifts more effectively, reducing long-term operational expenses. Additionally, short-term optimization strategies, such as preferring weekend or holiday shifts over weekday shifts or switching from a day shift to a night shift when energy prices are lower, can further reduce costs. With improved short-term electricity price forecasts, such adaptive scheduling decisions can be made more effectively. The results also suggest that assessing labor, energy, and inventory costs per shift can help managers identify cost-saving opportunities. By analyzing cost structures at a granular level, companies can refine their scheduling strategies to optimize resource utilization.

To fully leverage these opportunities, more flexible working time

Table 8
Total costs.

Total Costs	Scenario I	Scenario II	Scenario III
Capacity investment costs	0	0	4 146 227
Labor costs	13 943 899	14 798 355	14 129 022
Energy costs	31 601 212	23 315 501	10 366 521
Inventory costs	2 129 778	1 310 816	2 643 586
Overall Costs	**47 674 889**	**39 424 672**	**31 285 356**

models are essential. For example, instead of maintaining a fixed night shift schedule, companies could cancel the Thursday night shift and replace it with a Sunday night shift when electricity costs are lower. Such dynamic scheduling approaches would allow for further cost reductions while maintaining production efficiency.

In addition to cost considerations, companies must also address broader sustainability objectives. Strategies should aim not only to reduce energy consumption but also to lower carbon emissions, while maintaining the cost-effectiveness of other production factors. The use of renewable energy can support lower carbon emissions, although it may introduce additional variability in energy prices and availability. Therefore, managers must carefully assess energy-specific requirements, dynamic pricing constraints, workforce availability, logistics, and inventory limitations when developing production plans.

From a sustainability standpoint, flexible scheduling models must also consider the impact on employees. Night, weekend, and holiday shifts may pose challenges for workers, particularly those with families, and could affect health and safety if not managed carefully. Therefore, optimization strategies must balance economic efficiency with fair and sustainable working conditions.

Ultimately, a well-designed production and scheduling strategy should aim to achieve both short-term cost savings and long-term profitability, while also contributing to environmental and social sustainability goals. Jaehn (2016) emphasizes the inclusion of the social aspect of sustainability in production planning problems as an important area for future research.

7. Conclusion and limitations

The transformation of the energy market toward renewable energy and dynamic price structures offers the potential for energy-intensive industries to optimize production activities and reduce energy costs. This study presents an optimization model for production planning and decision-making on shift activation and machinery investment, aiming to minimize total costs related to inventory, energy, and labor. The model demonstrates how integrating shift-dependent cost factors for energy and labor can lead to improved decision-making. Unlike traditional production planning approaches that treat labor and energy costs as static, this model dynamically adjusts shift activation to minimize overall costs. The findings indicate that activating shifts with lower energy prices and labor costs significantly reduces total expenses while maintaining production efficiency. The results show that investment in additional machine capacity enhances flexibility in shift scheduling. While increased capacity leads to higher upfront investment costs, it allows for better alignment of production with cost-efficient shifts, ultimately reducing long-term operational expenses. This trade-off between investment and operational cost savings is a crucial consideration for decision-makers in energy-intensive industries. The applicability of the model was demonstrated through a practice-relevant use case involving multiple scenarios and a sensitivity analysis.

Despite its practical applicability, the model has certain limitations. It does not explicitly consider production complexities, such as setup times and production interruptions, which may affect real-world feasibility. Furthermore, it does not differentiate between energy sources, such as renewable and fossil fuels, which could influence both cost structures and shift availability. Additionally, while grouping similar products, machines, and personnel qualifications simplifies the model, it may overlook specific constraints in detailed production planning.

To enhance the model's applicability and robustness, future research should focus on validation through real-world implementation. Testing the model with actual production data would provide insights into its effectiveness and highlight necessary refinements. Additionally, extending the model to handle multiple production programs within an activated shift would make it more flexible for various industries. Another important research direction is the development of specialized solvers that consider the model's specific structure. For instance, a solver that ensures non-negative inventory by automatically activating the cheapest shift, when necessary, could improve computational efficiency and practical usability. By addressing these limitations and extending the model, future research can further bridge the gap between theoretical optimization and real-world industrial applications.

Declaration about generative AI

While preparing this work, the authors used Grammarly to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

CRediT authorship contribution statement

Herbert Jodlbauer: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Sonja Strasser:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Klaus Altendorfer:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Shailesh Tripathi:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Ann-Kristin Thienemann:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Nadine Bachmann:** Writing – review & editing, Writing – original draft, Validation, Formal analysis. **Manuel Brunner:** Writing – review & editing, Writing – original draft, Validation, Formal analysis. **Julian M. Müller:** Writing – review & editing, Writing – original draft, Validation, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Herbert Jodlbauer reports financial support was provided by Christian Doppler Research Association. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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